



Showrav Das

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ABOUT ME

I am a dedicated researcher with a strong foundation in **Data Science and Machine Learning**. My current research interest focuses on **Digital Twin-enabled Predictive Maintenance** for smart manufacturing systems. I am particularly interested in developing data-driven approaches for **health monitoring, fault diagnosis, and predictive analytics** of complex industrial systems.

With international academic exposure through the **Erasmus+ program**, I aim to further explore how **Digital Twin** technology can bridge physical systems and intelligent data models to improve reliability and maintenance strategies in modern industries.

EDUCATION AND TRAINING

01/03/2024 – 12/07/2024 Gliwice, Poland
ERASMUS+ KA107 INTERNATIONAL CREDIT MOBILITY Silesian University of Technology

Computer Networks, Fundamentals of Database Systems, Microprocessor and Embedded Systems, Operating Systems, Software Engineering

Website www.polsl.pl/en/ | **Level in EQF** EQF level 6 | **Type of credits** ECTS | **Number of credits** 15

10/01/2022 – 17/01/2026 Dhaka, Bangladesh
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING Daffodil International University

Programming and Problem Solving, Object-Oriented Programming, Data Structures, Algorithms, Discrete Mathematics, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Computer Architecture and Organization, Digital Electronics, Embedded Systems and IoT, Numerical Methods, Artificial Intelligence, Data Mining and Machine Learning, Compiler Design, Information Security, Web Engineering, Mobile Application Development, Computer Graphics, Natural Language Processing, Statistics and Probability

Website www.daffodilvarsity.edu.bd | **Final grade** 3.73 out of 4.00 | **Level in EQF** EQF level 6 |

Thesis Title: "Digital Twin-Enabled Physics-Constrained GAN for Li-ion Battery Health Prognostics." Developed a Digital Twin framework for the Predictive Maintenance of complex systems. I integrated domain-specific physics constraints with generative models to bridge the gap between physical assets and data-driven insights. By utilizing large-scale datasets, I optimized prognostic accuracy and reliability engineering strategies to drive innovation in intelligent, autonomous maintenance systems.

WORK EXPERIENCE

WORDPRESS DEVELOPER & LEAD GENERATION OFFICER – BLIZ IT – 01/07/2024 – 30/08/2025 – DHAKA, BANGLADESH

- **Developed and maintained** functional, responsive WordPress websites, ensuring an optimal user experience across devices.
- **Customized** themes and plugins while optimizing site performance and security to meet specific client requirements.
- Led lead generation campaigns to support client acquisition and business growth.
- Collaborated with clients and cross-functional teams to deliver tailored solutions.

SKILLS

WordPress | database | Data Science | Data Collection, Data Processing, Data Analysis, Data Visualisation | Data Science: Python (requests, selenium, BeautifulSoup, pandas, matplotlib, seaborn, sklearn) | Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow)

HONOURS AND AWARDS

23/01/2024
The Duke of Edinburgh International Award (Silver and bronze) – The Duke of Edinburgh’s International Award Foundation

- **Gold Award Participant (In Progress):** Currently completing advanced requirements in leadership, community service, and physical endurance.
- **Silver & Bronze Award Recipient:** Successfully completed sustained programs in **community service, leadership development, and outdoor expeditions**.
- **Key Competencies:** Developed strong resilience, cross-cultural teamwork, and problem-solving skills through structured personal and professional challenges.

VOLUNTEERING

01/07/2023 – 30/12/2023 International Affairs, Daffodil International University
Volunteer Internship program (Season-2)

- **Managed** administrative procedures for visa processing and delegate registration.
- **Organized** high-level international events, ensuring seamless coordination between organizers and foreign guests.
- **Developed** strong organizational and interpersonal skills in a multicultural environment.

PUBLICATIONS

Physics-Informed LSTM for State-of-Health Prediction of Li-ion Batteries Using the NASA Dataset (In Progress)

[Submitted to 3rd International Conference on Machine Intelligence and Emerging Technologies (MIET 2026)]

Completed a conference research paper proposing a Physics-Informed LSTM model for lithium-ion battery State-of-Health forecasting, integrating monotonicity and smoothness constraints to ensure physically consistent degradation prediction for predictive maintenance.

Authors: Showrav Das, Zadid Al Lisan, Md. Mehrab Hossain Badhan, Dr. Md Alamgir Kabir

Digital Twin-Enabled Physics-Constrained GAN for Li-ion Battery Health Prognostics [In Progress]

Developed a Digital Twin-based framework for lithium-ion battery health prognostics using a Physically Constrained Generative Adversarial Network (PCGAN). The model integrates physics-informed constraints with deep learning to generate physically consistent degradation trajectories and improve State-of-Health prediction using the NASA battery aging dataset. The work is being prepared for journal publication under the supervision of **Dr. Md. Alamgir Kabir**.

Authors: Showrav Das, Dr. Md Alamgir Kabir

SpongePath: A Dataset for Sponge Gourd Leaf Disease Classification

Completed a conference research paper on automated sponge gourd leaf disease identification using machine learning and deep learning approaches, including SVM, CNN, and VGG16 transfer learning. The study achieved up to **92% classification accuracy** and explores computer vision-based solutions for precision agriculture applications. The paper was previously accepted at **IDDA 2025** and is currently submitted to **3rd International Conference on Machine Intelligence and Emerging Technologies (MIET 2026)**.

Authors: Zadid Al Lisan, Showrav Das, Md Mehrab Hossain Badhan, Mayen Uddin Mojumdar, Md Alamgir Kabir

CERTIFICATIONS

Oracle Dev Gym, 10/04/2026

Databases for Developers: Foundations

I earned a 92% score, and the course was taught by Chris Saxon—packed with practical, hands-on learning. Here's what I worked through:

- Database tables, columns, and data types,
 - Data modeling fundamentals
 - Writing SQL queries using SELECT & WHERE,]
 - Working with different types of JOINS
 - Aggregations and GROUP BY
 - Data manipulation: INSERT, UPDATE, DELETE, TRUNCATE,
 - Transactions, COMMIT & ROLLBACK & description...

Mode of learning: Online

Universitas Surabaya (UBAYA), Office of International Affairs, 03/12/2023

Ubaya Online Summer Program 2023: "ChatGPT, Artificial Intelligence, & Metaverse for Our Daily Life"

- Explored the practical applications and integration of **Artificial Intelligence (AI)**, **ChatGPT**, and the **Metaverse** in contemporary daily life.
- Gained insights into digital transformation and emerging technologies within an international academic framework.

Mode of learning: Online

REFERENCES

Dr. Md Alamgir Kabir

Relation: Thesis & work Supervisor

Designation: Assistant professor & coordinator (Msc in Management Information Systems program)

Department: Computer Science & Engineering

Institution: Daffodil International University

Email: **kabir.cse@diu.edu.bd**

Dr. Arif Mahmud

Relation: Academic Supervisor

Designation: Associate Professor & Associate Head

Department: Computer Science & Engineering

Institution: Daffodil International University

Email: **arif.cse@diu.edu.bd**